Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 1 Quiz 1 - Kinematics in One Dimension

Mock Exam B · Form 2: Single-step application

Multiple Choice Questions: Select the best answer. (1 point each)

1. A cart is currently at x = -8 m with v = -5 m/s and a = 1 m/s². Which statement correctly describes its motion at this instant?

(A) It moves left and slows down.

(B) The object is currently moving to the right.

(C) The x–t slope is zero and the object is at rest.

(D) A negative velocity means a positive x–t slope.

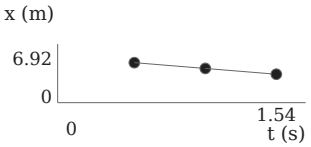
2. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.54, 6.92); (1.04, 5.92); (1.54, 4.92). What is the slope between the endpoint measurements?

(A) -2.8

(B) -0.4

(C) -1.2

(D) -2



3. A motion sensor records a cart starting at +3 m/s. Acceleration is +3 m/s² for 2 s, then 0 m/s² for another 2 s. What is its final velocity?

(A) 18 m/s

(B) 4.5 m/s

(C) 9 m/s

(D) -9 m/s

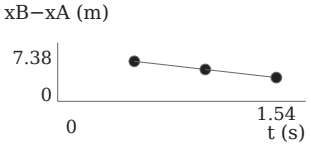
4. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.54, 7.38); (1.04, 5.88); (1.54, 4.38). Which relation models this apparatus?

(A) d=d0+(vB-vA)t

(B) Y=-6X+(9)

(C) Y=3X+(9)

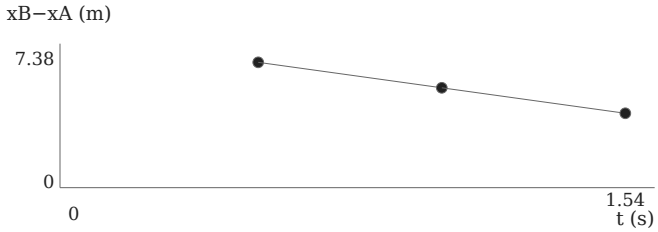
(D) Y=-3X²+(9)



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. A runs at +5 m/s and B at +2 m/s. Record the lead of B ahead of A as a signed separation. t (s) -> xB-xA (m): (0.54, 7.38); (1.04, 5.88); (1.54, 4.38).

a. For the runner separation setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Label the axes and units and demonstrate interpolation between two readings. **(3 pt)**

6. A cart brakes immediately at a constant magnitude of 2 m/s². Read its stopping coordinate along the track. u² (m²/s²) -> xstop (m): (1.08, 1.27); (2.08, 1.52); (3.08, 1.77).

a. Build the model for stopping-distance test from physics and predict both slope and intercept. **(2 pt)**

b. Label the axes and units and demonstrate interpolation between two readings. **(2 pt)**

7. Photogates record the position of a cart traveling at constant velocity. t (s) -> x (m): (0.54, 6.92); (1.04, 5.92); (1.54, 4.92). Build the model for track photogates from physics and predict both slope and intercept. **(3 pt)**